

# Irina Saranskaia

• irinasaranskaia@gmail.com •

Research interests: speech processing, predictive coding, neuroimaging analysis

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## Education

### HSE University

PhD in Cognitive Sciences

Moscow, Russia, 2025 – present

### HSE University

M.S., Cognitive Sciences and Technologies

Moscow, Russia, 2023 – 2025

Graduated with excellence

### St. Petersburg State University

B.S., Computer Science and Artificial Intelligence

St. Petersburg, Russia, 2019 – 2023

Graduated with excellence

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## Publications

Vlasenko, D., [Saranskaia, I.](#), & Zakharov, D. (2026). Hypergraphs from multivariate connectivity: caCoh-based EEG/MEG representation. [arXiv](#).

Vlasenko, D., [Saranskaia, I.](#), & Zakharov, D. (2025). From pairwise to high-order: Hypergraph methods and multivariate connectivity metrics for EEG/MEG. In *Proceedings of the 2025 7th International Conference Neurotechnologies and Neurointerfaces (CNN)*, 142–145, [doi.org/10.1109/CNN67635.2025.11177482](https://doi.org/10.1109/CNN67635.2025.11177482)

[Saranskaia, I.](#), Gutkin, B., & Zakharov, D. (2025) Aim-based choice of strategy for MEG-based brain state classification. *European Physical Journal Special Topics*, 234, 4159–4177. [doi.org/10.1140/epjs/s11734-025-01587-y](https://doi.org/10.1140/epjs/s11734-025-01587-y)

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## Research experience

### HSE University, Institute for Cognitive Neurosciences

#### Project "AI-based listening diagnostic systems"

Research assistant, February 2025 – present

- Full EEG experiment pipeline execution: experimental design, stimulus set construction, pilot validation, large-scale data acquisition
- Data quality control, trial validation, speech recordings analysis
- EEG preprocessing, time-frequency and spectral analyses, feature analysis
- Scientific writing

Technologies: Python: psycopy, mne, numpy, pandas, scipy, scikit-learn, seaborn and matplotlib, speech/behavioural tooling (JiWER for WER/CER; audio processing with librosa/pyloudnorm/parselmouth/soundfile), Hilbert/PSD/Welch methods; MATLAB: fieldtrip; Praat scripting and .TextGrid annotation

### Russian Science Foundation

#### Grant 24-68-00030 "Next Generation Cognitive Artificial Intelligence"

Research-assistant, December 2024 – present

- EEG/MEG data analysis and feature extraction for graph representation for a classification of brain states

Technologies: Python: numpy, scipy, scikit-learn, pandas, matplotlib, mne, mne-connectivity, jpype, infodynamics; MATLAB: fieldtrip.

### HSE University

2023 – 2025

- Master's research dedicated to the MEG-based functional connectivity analysis in anxiety patients utilising Time-Reversed Granger causality for functional connectivity assessment.

Technologies: MEG connectivity analysis, time-reversed Granger causality; python, numpy, pandas, scipy, mne, scikit-learn; MATLAB: eeglab; inverse modelling, LCMV beamforming, statistical analysis, multiple comparison corrections.

#### **St. Petersburg State University**

2022 – 2023

- Bachelor's thesis "Reproducibility Study of Different Methods For Brain Networks Comparisons." The thesis served the purpose of formulating recommendations for the use of various correction methods for multiple comparisons.

Technologies: coherency-based EEG analysis; python, numpy, pandas, scipy, multiply; graph-theory, statistical analysis, nonparametric permutation testing, multiple comparison corrections.

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## **Conferences**

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### **XI International Conference on Cognitive Science.**

*"Functional brain connectivity and anxiety: An MEG study in the context of decision-making"* (poster presentation)

*"Pupil size and response accuracy in perceptual decision-making in an ambiguous image classification"* (poster presentation)

Nizhny Novgorod, Russia. 24 – 28 August 2026

### **NeuroAI Meeting 2025.**

*"ML-based approaches for EEG/MEG classification"* (oral presentation).

Moscow, Russia. 24 – 25 November 2025

### **VII International Conference "Neurotechnologies and Neurointerfaces" as part of the V International Conference "Volga Neuroscience Meeting 2025".**

*"From Pairwise to High-Order: Hypergraph Methods and Multivariate Connectivity Metrics for EEG/MEG"* (poster presentation)

Nizhny Novgorod, Russia. 25 – 29 August 2025

### **XXXI Scientific School "Nonlinear Days-2025".**

*"Aim-based choice of strategy for MEG-based brain state classification"* (poster presentation).

Saratov, Russia. 26 – 30 May 2025

### **XXI Scientific School "Nonlinear Waves-2024".**

*"Methods for assessing functional connectivity based on electromagnetic data of brain activity"* (poster presentation).

Nizhny Novgorod, Russia. 5 – 11 November 2024

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## **Scholarships**

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Ostrogradski Fellowship, Ambassade de France en Fédération de Russie

September – December 2026

Brain Programme Fellowship, Research Center "IDEAS"

January 2026 – present

Academic Scholarship, HSE University

November 2025 – present

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## **Additional Education**

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Scientific School "Methods of psycho- and neurolinguistics", HSE University

St. Petersburg, Russia. 22–24 June 2026

XIII Summer Neurolinguistics School "Clinical applications of neurolinguistics in neurosurgery", Center for Language and Brain, HSE University

Moscow, Russia. 1–3 June 2026

Summer school in NeuroAI, Neuromatch Academy

Online. 14 – 25 July 2025

Summer school in Computational Neuroscience, Neuromatch Academy

Online. 8 – 26 July 2024

Project "Neural correlates of visual processing and action initiation in mouse

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## Languages

Russian (native), English (C1), French (A2)

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## Skills

### Programming

Python: numpy, pandas, scipy, matplotlib, seaborn, scikit-learn, pytorch, nilearn, networkx, mne, mne-connectivity, infodynamics;  
PsychoPy (including custom experiment scripting);  
MATLAB: fieldtrip, eeglab;  
algorithms and data structures;  
version control systems (GitHub);  
HTML, CSS, LaTeX.

### Data analysis and machine learning

classical data analysis;  
analysis of categorical data;  
network analysis;  
machine learning, deep learning;  
numpy, pandas, scipy, matplotlib, scikit-learn, pytorch.

### Neuroimaging data processing

Analysis and preprocessing of neurophysiological data, including EEG/MEG (ERP/ERF, time-frequency, connectivity analyses) and intracranial recordings (LFP and spike analysis from Neuropixels probes);

### Neuroimaging experiment

EEG (BrainVision), EyeTracking (EyeLink)

### Speech and audio processing

Analysis and preprocessing of speech data (Praat scripting for segmentation, annotation, and acoustic feature extraction);  
Python-based audio libraries for speech and sound analysis: pyloudnorm, librose, parselmouth, signal.

**Mathematics:** statistics, probability theory, graph theory, linear algebra, calculus.

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## References

### Dr. Denis Zakharov

Director, Theoretical Neuroscience Group, Centre for Cognition & Decision Making, HSE University, Moscow, Russia  
Leading Research Fellow, Centre for Cognition & Decision Making, HSE University, Moscow, Russia  
[dgzakharov@hse.ru](mailto:dgzakharov@hse.ru)

### Dr. Olesia Dogonasheva

Postdoctoral Researcher, Institut de l'Audition (Institut Pasteur / Inserm / CNRS / Université Paris Cité / AP-HP), IHU reConnect, Paris, France  
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